

Curriculum Vitae

Tianzhong Yang
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Education

PhD, The University of Texas Health Science Center at Houston, 2018
Biostatistics
Dissertation/Thesis Title: *Statistical Methods for gene-environment interactions and high-dimensional mediation analysis*
Advisor: Peng Wei

MPH, Boston University, Biostatistics and Epidemiology 2012

BS, Shanghai Jiao Tong University, Shanghai, China. 2010
Biotechnology

Academic Appointments

University of Minnesota
Medical School, Department of Pediatric: Adjunct Assistant Professor 2023 - Present

University of Minnesota
School of Public Health, Division of Biostatistics and Health Data Science: Assistant Professor 2020 - Present

University of Minnesota
School of Public Health, Division of Biostatistics: Postdoc Associate 2018 - 2020
Advisor: Wei Pan

The University of MD Anderson Cancer Center
Department of Biostatistics, Graduate Research Assistant 2016 - 2018

The University of Texas Health Science Center at Houston
Department of Biostatistics, Graduate Research Assistant 2014 - 2016

The University of Texas Health Science Center at Houston
Department of Biostatistics, Teaching Assistant 2013 - 2015

Other Professional Positions

Boston University
Clinical Epidemiology Research and Training Unit, Statistician 2011 - 2013

Philips Research North America
Summer Data Scientist Intern 2016

HONORS AND RECOGNITION

University of Minnesota

BHDS Thank-a-Mentor recognition, School of Public Health	2026
Rewarding Excellence and Productivity, School of Public Health	2023

External Sources

Julia's Legacy of Hope Scholar, St. Baldrick's Foundation	2024 - Present
Robert H. Bigelow Endowed Scholarship, The University of Texas Health Science Center at Houston	2017
Summer Institute Scholarship Award, University of Washington	2017
Allan R. Meyers Memorial Fellowship for the advancement of careers in disability, Boston University	2011
Merit Scholarship, Boston University	2010
People's Scholarship in China, Shanghai Jiao Tong University	2009

RESEARCH, SCHOLARSHIP, AND CREATIVE WORK

Grants, Contract, Awards Currently Funded: External Sources

- Award: Evaluating Germline genetic risk factors of Ewing Sarcoma onset and survival**
Role: ***Principal Investigator***
Principal Investigator: Yang, Tianzhong (University of Minnesota)
Sponsoring Organization: St. Baldrick's Foundation
Sponsoring Organization Reference ID: 1463960
Award Dates: July 1, 2024 - June 30, 2027
Total Award Amount: \$279,998
Effort: 60% salary support (Year 1-2), 20% salary support (Year 3, Reduction of this career award due to receipt of another grant according to the award guideline)
Purpose: To better understand the genetic causes of Ewing Sarcoma, a rare bone tumor occurs among adolescents and young adults.
- Award: Polygenic predictors of primary ovarian insufficiency in survivors of childhood cancer**
Role: ***Co-Principal Investigator***
Co-Project Investigators: Im, Cindy (University of Minnesota)
Sponsoring Organization: The Children's Cancer Research Fund
Award Dates: August 2024 - July 2027
Total Award Amount: \$100,000
Effort: 1% salary support (Year 1), 5% salary support (Year 2)
Purpose: Evaluate known genetic risk factors and identify novel genetic predictors for menopausal timing and primary ovarian insufficiency among childhood cancer survivors. Adapt proven machine learning techniques (e.g., transfer learning methods) to leverage results from general population and survivor-specific GWAS to improve risk prediction among survivors.

3. **Award: Estimation and inference in directed acyclic graphical models for biological networks**

Role: Co-investigator

Principal Investigator: Shen, Xiaotong T. (University of Minnesota)

Sponsoring Organization: NIH National Institute on Aging (NIA)

Sponsoring Organization Reference ID: 1R01AG074858-01

Award Dates: April 1, 2022 - January 31, 2027

Total Award Amount: \$3,189,178

Effort: 15% salary support

Purpose: The proposed project will address fundamental issues for reconstruction and inference of high-dimensional causal networks, then apply the developed new methodology to the elucidation of causal genetic variants, genes, and risk factors for Alzheimer's disease (AD) with potential applications to other common illnesses, thus it will facilitate their prevention, early diagnosis, and therapeutic development.

4. **Award: Novel Methods to Quantify High-dimensional Omics Mediation Effects from Alcohol Intake to Coronary Heart Disease**

Role: ***Co-Principal Investigator***

Co-Project Investigators: Li, Chunlin (University of Virginia)

Sponsoring Organization: NIH National Heart, Lung, and Blood Institute

Award Dates: Jun 22, 2026 – Feb 28, 2030

Percent Effort: 30% salary support

Total Proposed Amount: \$3,259,742

Purpose: Propose new high-dimensional mediation analysis methods to leverage information across correlated traits, existing large-scale studies and multi-omics mediator. The project will quantify the missing mediation effects and shared mediation effect from weak signal strengths from well-established risk factors to cardiovascular diseases.

Previously Funded and Concluded

1. **Award: Sex and racial/ethnic differences in B-ALL genomics**

Role: Co-investigator

Principal Investigators: Spector, Logan G. (University of Minnesota)

Sponsoring Organization: NIH National Cancer Institute (NCI)

Award Dates: August 2024 - July 2025

Total Award Amount: \$155,000

Effort: 1% salary support

Purpose: To conduct sex specific genome-wide association studies using the same genome-wide array data being generating by the parent R01, and to identify sex-differential expression quantitative trait loci using a newly generated source of pediatrics data for African American children.

2. **Award: Genomics of childhood leukemia: Analysis of diverse, worldwide populations**

Role: Statistical Geneticists

Principal Investigator: Spector, Logan G. (University of Minnesota)

Sponsoring Organization: Blood Cancer United

Sponsoring Organization Reference ID: MCG-23020-22

Award Dates: March 1, 2022 - February 28, 2025

Total Award Amount: \$250,000

Effort: 3% salary support

Purpose: To aggregate, clean, and harmonize existing childhood cancer germline genotype data at three analysis centers.

3. **Award: Evaluating genetic risk factors of hepatoblastoma onset and survival**

Role: *Principal Investigator*

Sponsoring Organization: St. Baldrick's Foundation

Sponsoring Organization Reference ID: 932143

Award Dates: July 1, 2022 - June 30, 2024

Total Award Amount: \$229,998

Effort: 60% salary support

Purpose: To better understand the genetic causes of hepatoblastoma, a rare childhood liver cancer.

4. **Award: Somatic mosaicism in twins discordant for childhood cancer**

Role: Co-investigator

Principal Investigator: Poynter, Jenny N. (University of Minnesota)

Sponsoring Organization: Children's Cancer Research Fund

Award Dates: August 2022 - July 2023

Total Award Amount: \$100,000

Effort: 5% salary support (Year 1), 1% salary support (Year 2)

Purpose: Establish the feasibility of recruiting and conducting in home sample collection of biospecimens from same sex twins discordant for childhood CNS tumors; Compare the prevalence of somatic mosaicism in classic cancer-associated genes in same-sex twins discordant for childhood brain tumors; Compare profile of mutation detected in DNA extracted from pre- and post-treatment blood samples from CNS tumor patients

5. **Award: Evaluating non-genetic risk factors for hepatoblastoma through genetic epidemiology methods**

Role: *Principal Investigator*

Sponsoring Organization: Children's Cancer Research Fund

Award Dates: August 2021 - July 2023

Total Award Amount: \$100,000

Effort: 5% salary support

Purpose: Develop a new statistical agnostic screen method and apply it to search for non-genetic risk factors for hepatoblastoma; Verify the causality of putative non-genetic risk factors of HB using genetic epidemiological methods.

6. **Award: Admixture analysis of acute lymphoblastic leukemia in African American children: the ADMIRAL Study**

Role: Co-investigator

Principal Investigator: Spector, Logan G. (University of Minnesota)

Sponsoring Organization: NIH National Cancer Institute

Sponsoring Organization Reference ID: 1R01CA239701-01A1S1, S2 and S3

Award Dates: August 1, 2020 - July 31, 2025

Total Award Amount: \$3,838,319

Effort: 1% salary support

Purpose: To conduct admixture mapping of acute lymphoblastic leukemia risk and outcome in African American patients.

Pending/Submitted

1. **Proposal: Transcriptome-wide association study of childhood acute lymphoblastic leukemia informed by single-cell analysis of cord blood derived precursor cells**
Role: *Co-Principal Investigator*
Co-Project Investigators: Spector, Logan G. (University of Minnesota)
Sponsoring Organization: NIH National Cancer Institute
Date Submitted: June 5, 2026
Total Proposed Amount: \$3,960,762
Percent Effort: 30%
Purpose: Acute lymphoblastic leukemia (ALL) is the most common cancer in children. Common genetic variants related to white blood cell development have been associated with ALL. We will perform single-cell sequencing on cord blood data on African American kids and integrate the single cell data with childhood acute lymphoblastic leukemia for gene and cell-of-origin discovery.
2. **Proposal: Multi-ancestry maternal genomics and offspring risk of childhood acute lymphoblastic leukemia: the MaternALL study**
Role: *Co-Principal Investigator*
Co-Project Investigators: Spector, Logan G. (University of Minnesota)
Sponsoring Organization: NIH National Cancer Institute
Date Submitted: June 5, 2026
Percent Effort: 35%
Total Proposed Amount: \$4,464,827
Purpose: Create and analyze a multi-ancestry dataset of > 2800 genotyped mother-case to examine maternal genetics and offspring acute lymphoblastic leukemia.
3. **Proposal: Development of Tissue-Specific Toxicity Index (TSTI) for Human-Centric Drug Toxicity Prediction**
Role: Co-Investigator
Principal Investigator: Cheng, Shen (University of Minnesota)
Sponsoring Organization: NIH National Institute of General Medical Science
Date Submitted: June 3, 2026
Percent Effort: 10%
Total Proposed Amount: \$2,866,763
Purpose: Drug-induced toxicity prevents patients from receiving efficacious treatment and is a major reason for new drug development failure. This project will develop a novel computational drug toxicity prediction tool which can forecast the potential toxicity from thousands of drugs in a wide array of human tissues.
4. **Proposal: Statistical and Computational Methods for Policy-Aware Federated Genomics**
Role: Co-Investigator
Principal Investigator: Basu, Saonli (University of Minnesota)
Sponsoring Organization: NIH National Human Genome Research Institute
Date Submitted: February 11, 2026
Percent Effort: 10%
Total Proposed Amount: \$423,500
Status: Waiting for review
Purpose: This project will develop new statistical methods that enable powerful, privacy-preserving genomic analyses across major resources such as All of Us, UK Biobank, and local genomic data commons without sharing individual-level data. By improving how genetic information is safely combined across diverse populations, this work will advance

equitable and reproducible genomic research with direct relevance to public.

5. **Proposal: Genomic Insights into Immune Aging: Exploring the Impact of cytomegalovirus infection on T-Cell Dynamics**

Role: *Co-Principal Investigator*

Co-Project Investigators: Guan, Weihua and Thyagarajan, Bharat (University of Minnesota)

Sponsoring Organization: NIH National Institute on Aging

Date Submitted: February 4, 2026

Percent Effort: 20%

Total Proposed Amount: \$3,498,142

Status: Waiting for review

Purpose: Understand the genetic mechanisms driving T-cell aging, particularly the influence of chronic infections due to CMV.

6. **Proposal: Early Statin Therapy for Asymptomatic Osteonecrosis After Acute Lymphoblastic Leukemia: A Controlled Pilot Study of Feasibility, Safety, and Preliminary Efficacy Signals**

Role: Co-Investigator

Principal Investigator: Cheng, Edward (University of Minnesota)

Date Submitted: March 16, 2026

Percent Effort: 5%

Total Proposed amount: \$421,954

Purpose: Determine the effect of statin therapy on preventing osteonecrosis in adolescent and young adult patients with ALL; Determine the potential impact of statin therapy on leukemia-related outcomes in adolescents and young adults with ALL; Assess the safety and tolerability of statin therapy in the context of ALL treatment.

Grants, Contract, Awards Currently Funded: Internal Sources

Previously Funded and Concluded

1. **Award: Statistical Models for Multi-Omics Multi-Cell Deconvolution**

Role: *Co-Principal Investigator*

Co-Project Investigators: Guan, Weihua (University of Minnesota)

Sponsoring Organization: Division of Biostatistics and Health Data Science at University of Minnesota (BHDS Seed Grant)

Award Dates: August 2022 - August 2024

Total Award Amount: \$109,836

Effort: 0% salary support

Purpose: To develop novel statistical models for cell deconvolution for bulk tissue

Pending/Submitted

1. **Proposal: Integrative genomics study of pediatric CNS cancer using Healthy Brain and Child Development and Molecular Characterization Initiative studies**

Role: *Co-investigator*

Principal Investigator: Spector, Logan G. (University of Minnesota)

Sponsoring Organization: Masonic Institute for the Developing Brain at University of Minnesota

Date Submitted: June 24, 2026

Total Proposed Amount: \$100,000

Percent Effort: 0%

Purpose: To perform QTL analyses in HBCD data and prioritize QTL in relation to specific molecular subtypes of CNS cancer; To perform GWAS analysis and mutational signature analysis from tumor sequences for further subtyping.

Publications

Hash(#) - indicates co-first author

Underline - indicates student author

Peer-Reviewed Journal Article

1. Kang, Z., Chen, L.[#], Wei, P., Xu, Z., Li, C., & **Yang, T.** (2026). Estimation of total mediation effect for a binary trait in a case-control study for high-dimensional omics mediators. *Annals of Applied Statistics* [In Press] PubMed Central ID Number: PMC11838279 [doi: 10.1101/2025.01.28.635396](https://doi.org/10.1101/2025.01.28.635396)
2. Arafa, A., Boytim, E., **Yang, T.**, & Antonarakis, E. S. (2026). Plasma Extracellular-Vesicle Proteomics Nominates Candidate Biomarkers of ¹⁷⁷Lu-PSMA-617 Outcomes in Metastatic Prostate Cancer Patients. *Cell Reports Medicine*, 7(5), 102764. PubMed Central ID Number: PMC13198300 [doi: 10.1016/j.xcrm.2026.102764](https://doi.org/10.1016/j.xcrm.2026.102764)
3. Cao, R., Li, C., Cui, C., Spector, L., Raduski, A., Anderson, N., . . . **Yang, T.** (2026). A Pseudotime-dependent TWAS framework identifies disease genes along cell developmental paths. *HGG Advances*. [Accepted] PubMed Central ID Number: PMC12636649 [doi: 2025.10.25338929](https://doi.org/2025.10.25338929)
4. Im, C., Raduski, A. R., Mills, L. J., Bhattarai, K. R., Mobley, R. J., Barnett, K. R., Lu, Z., Liao, K., Anderson, N., Johnson, R. A., Langer, E., Hooten, A. J., Seif, A. E., Bernt, K. M., Tsang, M., Mamou, B. A., Gil-de-Gómez, L., Wolfson, J. A., Friedman, D. N., Shukla, N., Klesse, L. J., Marcotte, E. L., Ji, L., Dang, A., Luo, M., Zhong, Y., Langie, J., Chiang, C. W. K., de Smith, A., Wiemels, J. L., DeWan, A., Ma, X., Metayer, C., Wang, Z., Nelson, H. H., Pankratz, N., **Yang, T.**, . . . Spector, L. G. (2025). Genome-wide association study of childhood B-cell acute lymphoblastic leukemia reveals novel African ancestry-specific susceptibility loci. *Nature communications*, 16(1), 8974. PubMed Central ID Number: PMC12546916 [doi: 10.1038/s41467-025-64337-7](https://doi.org/10.1038/s41467-025-64337-7)
5. Xie, T., Sorenson, J. C.[#], Spector, L. G., Pankratz, N., Huang, R. S., Hiyama, E., . . . **Yang, T.** (2025). Multiancestry Transcriptome-Wide Association Study Identifies Candidate Genes Associated with Hepatoblastoma. *Cancer epidemiology, biomarkers & prevention : a publication of the American Association for Cancer Research, cosponsored by the American Society of Preventive Oncology*, 34(8), 1405-1414. [doi: 10.1158/1055-9965.EPI-24-1553](https://doi.org/10.1158/1055-9965.EPI-24-1553)
6. Xu, Z., Li, C., Chi, S., **Yang, T.**, & Wei, P. (2024). Speeding up interval estimation for R2-based mediation effect of high-dimensional mediators via cross-fitting. *Biostatistics (Oxford, England)*, 26(1). PubMed Central ID Number: PMC11823199 [doi: 10.1093/biostatistics/kxae037](https://doi.org/10.1093/biostatistics/kxae037)
Role: Conceptualized the idea, supervised the project, worked closely with the student researcher, and critically revised the manuscript
7. Lucas, S. E., **Yang, T.**, Wimberly, C. E., Parmar, K. V., Hansen, H. M., de Smith, A. J., . . . Walsh, K. M. (2024). Genetic variation near GRB10 associated with bone growth and osteosarcoma risk in canine and human populations. *Cancer epidemiology*, 92, 102599. PubMed Central ID Number: PMC11402579 [doi: 10.1016/j.canep.2024.102599](https://doi.org/10.1016/j.canep.2024.102599)
Role: Conducted some of the statistical analyses, wrote part of the statistical methods/results, and critically revised the manuscript
8. Gianferante, D. M., Moore, A., Spector, L. G., Wheeler, W., **Yang, T.**, Hubbard, A., . . . Mirabello, L. (2024). Genetically inferred birthweight, height, and puberty timing and risk of osteosarcoma. *Cancer epidemiology*, 92, 102432. PubMed Central ID Number: PMC10869637 [doi: 10.1016/j.canep.2024.102432](https://doi.org/10.1016/j.canep.2024.102432)

- [10.1016/j.canep.2023.102432](https://doi.org/10.1016/j.canep.2023.102432)
9. Im, C., Neupane, A., Baedke, J. L., Lenny, B., Delaney, A., Dixon, S. B., Chow, E. J., Mostoufi-Moab, S., **Yang, T.**, . . . Sapkota, Y. (2024). Trans-Ancestral Genetic Risk Factors for Treatment-Related Type 2 Diabetes Mellitus in Survivors of Childhood Cancer. *Journal of clinical oncology : official journal of the American Society of Clinical Oncology*, 42(19), 2306-2316. PubMed Central ID Number: PMC11209771 [doi: 10.1200/JCO.23.02281](https://doi.org/10.1200/JCO.23.02281)
 10. Zhu, H., Choi, J., Kui, N., **Yang, T.**, Wei, P., Li, D., & Sun, R. (2024). Identification of Pancreatic Cancer Germline Risk Variants With Effects That Are Modified by Smoking. *JCO precision oncology*, 8, e2300355. PubMed Central ID Number: PMC11000774 [doi: 10.1200/PO.23.00355](https://doi.org/10.1200/PO.23.00355)
 11. **Zilinskas, R.**, **Li, C.**, Shen, X., Pan, W., & **Yang, T.** (2024). Inferring a directed acyclic graph of phenotypes from GWAS summary statistics. *Biometrics*, 80(1). PubMed Central ID Number: PMC10928990 [doi: 10.1093/biometc/ujad039](https://doi.org/10.1093/biometc/ujad039)
 12. **Yang, T.**, Mills, L. J., Hubbard, A. K., **Cao, R.**, Raduski, A., Machiela, M. J., & Spector, L. G. (2024). Genetic analyses identify evidence for a causal relationship between Ewing sarcoma and hernias. *HGG advances*, 5(1), 100254. PubMed Central ID Number: PMC10692953 [doi: 10.1016/j.xhgg.2023.100254](https://doi.org/10.1016/j.xhgg.2023.100254)
 13. **Cao, R.**, Olawsky, E., McFowland, 3rd, E., Marcotte, E., Spector, L., & **Yang, T.** (2024). Subset scanning for multi-trait analysis using GWAS summary statistics. *Bioinformatics (Oxford, England)*, 40(1). PubMed Central ID Number: PMC11087659 [doi: 10.1093/bioinformatics/btad777](https://doi.org/10.1093/bioinformatics/btad777)
 14. **Yang, T.**, Mills, L. J., Xue, H., Raduski, A., Williams, L. A., & Spector, L. G. (2022). Impact of fetal expression quantitative trait loci on transcriptome-wide association study of childhood leukemia. *Human molecular genetics*, 31(19), 3207-3215. [doi: 10.1093/hmg/ddab336](https://doi.org/10.1093/hmg/ddab336)
 15. Cigan, S. S., Meredith, J. J., Kelley, A. C., **Yang, T.**, Langer, E. K., Hooten, A. J., . . . Poynter, J. N. (2022). Predicted leukocyte telomere length and risk of germ cell tumours. *British journal of cancer*, 127(2), 301-312. PubMed Central ID Number: PMC9296514 [doi: 10.1038/s41416-022-01798-3](https://doi.org/10.1038/s41416-022-01798-3)
 16. **Yang, T.**, Jackson, V. E., Smith, A. V., Chen, H., Bartz, T. M., Sitlani, C. M., . . . Morrison, A. C. (2021). Rare and low-frequency exonic variants and gene-by-smoking interactions in pulmonary function. *Scientific reports*, 11(1), 19365. PubMed Central ID Number: PMC8481467 [doi: 10.1038/s41598-021-98120-7](https://doi.org/10.1038/s41598-021-98120-7)
 17. **Yang, T.**, Niu, J., Chen, H., & Wei, P. (2021). Estimation of total mediation effect for high-dimensional omics mediators. *BMC bioinformatics*, 22(1), 414. PubMed Central ID Number: PMC8381496 [doi: 10.1186/s12859-021-04322-1](https://doi.org/10.1186/s12859-021-04322-1)
 18. **Yang, T.**, Wei, P., & Pan, W. (2021). Integrative analysis of multi-omics data for discovering low-frequency variants associated with low-density lipoprotein cholesterol levels. *Bioinformatics (Oxford, England)*, 36(21), 5223-5228. PubMed Central ID Number: PMC7850048 [doi: 10.1093/bioinformatics/btaa898](https://doi.org/10.1093/bioinformatics/btaa898)
 19. **Yang, T.**, Tang, H., Risch, H. A., Olson, S. H., Peterson, G., Bracci, P. M., . . . Wei, P. (2020). Incorporating multiple sets of eQTL weights into gene-by-environment interaction analysis identifies novel susceptibility loci for pancreatic cancer. *Genetic epidemiology*, 44(8), 880-892. PubMed Central ID Number: PMC7657998 [doi: 10.1002/gepi.22348](https://doi.org/10.1002/gepi.22348)
 20. **Yang, T.**, Wu, C., Wei, P., & Pan, W. (2020). Integrating DNA sequencing and transcriptomic data for association analyses of low-frequency variants and lipid traits. *Human molecular genetics*, 29(3), 515-526. PubMed Central ID Number: PMC7015848 [doi: 10.1093/hmg/ddz314](https://doi.org/10.1093/hmg/ddz314)
 21. **Yang, T.**, Kim, J., Wu, C., Ma, Y., Wei, P., & Pan, W. (2020). An adaptive test for meta-analysis of rare variant association studies. *Genetic epidemiology*, 44(1), 104-116. PubMed Central ID Number:

PMC6980317 [doi: 10.1002/gepi.22273](https://doi.org/10.1002/gepi.22273)

22. **Yang, T.**, Chen, H., Tang, H., Li, D., & Wei, P. (2019). A powerful and data-adaptive test for rare-variant-based gene-environment interaction analysis. *Statistics in medicine*, 38(7), 1230-1244. PubMed Central ID Number: PMC6399020 [doi: 10.1002/sim.8037](https://doi.org/10.1002/sim.8037)
23. **Yang, T.**, Yang, Y., Jia, Y., & Li, X. (2019). Dynamic prediction of hospital admission with medical claim data. *BMC medical informatics and decision making*, 19(Suppl 1), 18. PubMed Central ID Number: PMC6354329 [doi: 10.1186/s12911-019-0734-y](https://doi.org/10.1186/s12911-019-0734-y)
24. Simon, D. K., Wu, C., Tilley, B. C., Lohmann, K., Klein, C., Payami, H., . . . **Yang, T.** (2017). Caffeine, creatine, GRIN2A and Parkinson's disease progression. *Journal of the neurological sciences*, 375, 355-359. PubMed Central ID Number: PMC5386398 [doi: 10.1016/j.jns.2017.02.032](https://doi.org/10.1016/j.jns.2017.02.032)
25. Guermazi, A., Eckstein, F., Hayashi, D., Roemer, F. W., Wirth, W., **Yang, T.**, . . . Felson, D. T. (2015). Baseline radiographic osteoarthritis and semi-quantitatively assessed meniscal damage and extrusion and cartilage damage on MRI is related to quantitatively defined cartilage thickness loss in knee osteoarthritis: the Multicenter Osteoarthritis Study. *Osteoarthritis and cartilage*, 23(12), 2191-2198. PubMed Central ID Number: PMC4957527 [doi: 10.1016/j.joca.2015.06.017](https://doi.org/10.1016/j.joca.2015.06.017)
26. Haugen, I. K., Ramachandran, V. S., Misra, D., Neogi, T., Niu, J., **Yang, T.**, . . . Felson, D. T. (2015). Hand osteoarthritis in relation to mortality and incidence of cardiovascular disease: data from the Framingham heart study. *Annals of the rheumatic diseases*, 74(1), 74-81. PubMed Central ID Number: PMC3959628 [doi: 10.1136/annrheumdis-2013-203789](https://doi.org/10.1136/annrheumdis-2013-203789)
27. Glass, N. A., Torner, J. C., Frey Law, L. A., Wang, K., **Yang, T.**, Nevitt, M. C., . . . Segal, N. A. (2013). The relationship between quadriceps muscle weakness and worsening of knee pain in the MOST cohort: a 5-year longitudinal study. *Osteoarthritis and cartilage*, 21(9), 1154-9. PubMed Central ID Number: PMC3774035 [doi: 10.1016/j.joca.2013.05.016](https://doi.org/10.1016/j.joca.2013.05.016)
28. Felson, D. T., Niu, J., **Yang, T.**, Torner, J., Lewis, C. E., Aliabadi, P., . . . Nevitt, M. C. (2013). Physical activity, alignment and knee osteoarthritis: data from MOST and the OAI. *Osteoarthritis and cartilage*, 21(6), 789-95. PubMed Central ID Number: PMC3648587 [doi: 10.1016/j.joca.2013.03.001](https://doi.org/10.1016/j.joca.2013.03.001)
29. Roemer, F. W., Felson, D. T., **Yang, T.**, Niu, J., Crema, M. D., Englund, M., . . . Guermazi, A. (2013). The association between meniscal damage of the posterior horns and localized posterior synovitis detected on T1-weighted contrast-enhanced MRI--the MOST study. *Seminars in arthritis and rheumatism*, 42(6), 573-81. PubMed Central ID Number: PMC3640766 [doi: 10.1016/j.semarthrit.2012.10.005](https://doi.org/10.1016/j.semarthrit.2012.10.005)

Conference Proceeding

1. Jia, Y., Wang, X., Cao, H., Ru, B., **Yang, T.** (2018) An empirical study of using an ensemble model in e-Commerce Taxonomy Classification Challenge. SIGIR eCom 2018 Data Challenge Papers.
Role: Statistical consulting and critically revised the manuscript

Software

1. Cao, R., **Yang, T.** TraitScan: *A Trait Subset Scan Method for Multi-trait GWAS*. R package version 0.1.0.
<https://github.com/RuiCao34/TraitScan>
2. Li, C., **Yang, T.**, Zilinskas, R., Shen, X., Pan, W. (2023) *sumDAG: Estimation and inference of directed acyclic graph from GWAS summary statistics*. R package version 0.9.0.
<https://github.com/chunlinli/sumdag>
3. **Yang, T.** (2023) *RsqMed: Total Mediation Effect Size Measure for High-Dimensional Mediators*. R

package version 1.1. <https://cran.r-project.org/web/packages/RsqMed>

4. **Yang, T.** (2018) *aGE: Adaptive Set-based Gene-Environmental Interaction Test for Rare Variants*. R package version 0.2. <https://github.com/ytzhong/aGE>
5. **Yang, T.** (2018) *metaRV: An adaptive test for meta-analysis of rare variants in sequencing studies*. R package version 1.0. <https://github.com/ytzhong/metaRV>

Patents and Intellectual Property

1. Huang, S.R., Cheng, S., **Yang, T.**, Shan, Y., Wang, L., Chaudhari, A. Tissue Specific Drug Induced Toxicity Forecaster. (Under filing at University of Minnesota, 2026)
2. Yang, Y., **Yang, T.**, Sedsh R., Jia Y. Systems and methods for dynamic monitoring of patient conditions and prediction of adverse events. Issued: May 14, 2020

Invited Presentations, Posters, and Exhibits

Presentation/Talk

1. **Yang, T.** "Evaluating General Population Polygenic Risk Score in Childhood Cancer Survivors for Anthropometric Traits," RR-BIG seminar University of Minnesota. (July 2026).
2. **Yang, T.** "Causal Models for High-dimensional Genomics Data," University of Tennessee Health Science Center, Virtual. (February 2026).
3. **Yang, T.** "Quantifying the Total Mediation Effect for Nonsparse High-dimensional Genomics Mediators," Joint Statistical Meeting, Nashville, Tennessee. (August 2025).
4. **Yang, T.** "Quantifying the Total Mediation Effect for Nonsparse High-dimensional Genomics Mediators," International Chinese Statistical Association Applied Statistics Symposium International Chinese Statistical Association, Storrs, Connecticut. (June 2025).
5. **Yang, T.** "Quantifying the Total Mediation Effect for Nonsparse High-dimensional Genomics Mediators," A Special Conference on Biostatistics in the New Era of Data-Driven Sciences, Houston, Texas, United States. (November 2024).
6. **Yang, T.** "Statistical Models for multi-trait analysis using summary-level genetic data," Medical College of Wisconsin, Milwaukee, Wisconsin, United States. (April 2024).
7. **Yang, T.** "Statistical Models for Advancing Childhood Cancer Research with Genomics Data," University of Minnesota, Minneapolis, Minnesota. (March 2024).
8. **Yang, T.** "Inferring a directed acyclic graph of phenotypes from GWAS summary statistics," CMStatistics. (December 2023).
9. **Yang, T.** "Subset scanning for multi-trait analysis using GWAS summary statistics," University of Mississippi Medical Center. (September 2023).
10. **Yang, T.** "Subset scanning for multi-trait analysis using GWAS summary statistics," Joint Statistical Meeting, Toronto, Canada. (August 2023).
11. **Yang, T.** "Subset scanning for multi-trait analysis using GWAS summary statistics," International Chinese Statistical Association Applied Statistics Symposium International Chinese Statistical Association, Ann Arbor, Michigan. (June 2023).

12. **Yang, T.** "Evaluate non-genetic risk factors for childhood cancers through genetic epidemiology methods," North Carolina at Greensboro, Greensboro, North Carolina, United States. (October 21, 2022).

Panelist

1. **Yang, T.** "Center for Innovative Design and Analysis Panel Discussion on Career experience," University of Colorado Anschutz. (April 2023). *Invited*.

Contributed presentations, Posters, and Exhibits

Underline - indicates student presenter

Poster

1. Sorenson, J., Raduski, A., Mills, L., Williams, L., Marcotte, E., Huang, R. H., Spector, L., **Yang, T.** "Exploring survival-associated tumor expression quantitative trait loci (eQTLs) in pediatric hepatoblastoma: Insights from a Japanese cohort," International Genetic Epidemiology Society Annual Meeting, Denver. (November 2024).
2. Cao, R., Olawsky, E., McFowland III, E., Marcotte, E., Spector, L., **Yang, T.** "TraitScan: subset scanning for multi-trait analysis using GWAS summary statistics," American Society of Human Genetics Annual Meeting, Washington. (November 2023).
3. Cao, R., Olawsky, E., McFowland III, E., Marcotte, E., Spector, L., **Yang, T.** "TraitScan: subset scanning for multi-trait analysis using GWAS summary statistics," Masonic Cancer Center. (October 2023).
4. Cao, R., Li, C., Cui, E., Spector, L., Im, C., Gordon, P., Raduski, A., Anderson, N., **Yang, T.** "A pseudo-time dependent TWAS framework identifies phenotype-associated genes along cell developmental paths.," STATGEN2025, Minneapolis, Minnesota. (May 2025).
5. Liu, Y., Im, C., **Yang, T.** "Improving polygenic risk prediction with pharmacogenomics: gene-radiotherapy interactions and impaired height in childhood cancer survivors," American Society of Human Genetics Annual meeting, Boston, Massachusetts, United States. (October 2025).
6. **Yang, T.** "Genomic analysis of Ewing Sarcoma and Hernia," Childhood Cancer and Leukemia International Consortium (CLIC) meeting, Seattle, Washington. (October 2023).

Presentation/Talk

1. Sirivivasan, A., Wang, J., Matt, G., Sapkota, Y., Wang, Z., **Yang, T.**, Neglia, J. P., Turcotte, L. M., Hudson, M. M., Ness, K. K., Armstrong, G. T., Zhang, J., Robison, L. L., Zhou, X., Yasui, Y., Im, C. "Clinical utility of 99 breast cancer polygenic risk scores in female survivors of childhood cancer: a CCSS and SJLIFE report," International Symposium on Late Complications after Childhood Cancer, Seattle, Washington, United States. (June 2025).
2. Sorenson, J., Raduski, A., Mills, L., Williams, L., Marcotte, E., Huang, R. H., Spector, L., **Yang, T.** "Exploring survival-associated tumor expression quantitative trait loci (eQTLs) in pediatric hepatoblastoma: Insights from a Japanese cohort," STATGEN 2025, Minneapolis, Minnesota. (May 2025).
3. Kang, Z., Wei, P., Xu, Z., Li, C., **Yang, T.** "Estimation of Total Mediation Effect for High-Dimensional Mediators in a Case-Control Study," ENAR, New Orleans. (March 2025).

4. **Yang, T.** "Subset scanning for multi-trait analysis using GWAS summary statistics," Joint statistical meetings, Toronto. (August 2023).
5. **Yang, T.** "Adaptive test for meta-analysis of rare variant association studies," International Genetic Epidemiology Society (IGES), Houston, Texas. (October 2019).
6. **Yang, T.** "Integrating DNA sequencing and transcriptomic data for association analyses of rare variants and lipid traits," American Society of Human Genetics (ASHG), Houston, Texas. (October 2019).
7. **Yang, T.** "Estimation of Mediation Effect for High-dimensional Omics Mediators with Application to the Framingham Heart Study," Joint Statistical Meeting (JSM), Denver, Colorado. (August 2019).
8. **Yang, T.** "Dynamic prediction of hospital admission with medical claim data," International Conference on Intelligent Biology and Medicine (ICIBM), Los Angeles, California. (June 2018).
9. **Yang, T.** "A Powerful and Data-Adaptive Test for Rare Variant-Based GxE Analysis," Eastern North American Region (ENAR), Atlanta, Georgia. (March 2018).
10. **Yang, T.** "A Powerful and Data-Adaptive Test for Rare Variant-Based GxE Analysis," JSM, Baltimore, Maryland. (August 2017).

TEACHING

Scheduled Teaching

PubH 8446: Advanced Topics in Omics Analysis	Spring 2026
PubH 6450: Biostatistics I (Co-taught with Laura Le)	Fall 2021 & 2023

Instructional Activity

University of Minnesota

1. Guest Lecture for Research Skills in Biostatistics, PUBH 8403
Topic: Choosing a dissertation topic and advisor
Fall 2021- 2022
2. Guest Lecture for Research Skills in Biostatistics, PUBH 8403
Topic: Formulating a research question and reviewing the literature
Fall 2025
3. Guest Lectures for Directed Research in Biostatistics, PUBH 8494
Co-led with Saonli Basu on organizing discussions and topics related to omics data analysis
Fall 2023
4. Guest Lectures for Biostatistics Modeling and Methods, PUBH 7402
An introduction to causal inference
Spring 2019

CURRICULUM DEVELOPMENT ACTIVITIES

Curriculum Development Activities

University of Minnesota

Revised and updated a new course curriculum, PUBH 8446 Advanced Topics in Omics Data Analysis

Spring 2026

ADVISING AND MENTORING

Graduate Student Activities

Academic Advisees

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| 1. Joshua Hebeisen, Biostatistics MS | 2025- Present |
| 2. Josey Sorenson, Biostatistics PhD | 2025- Present |
| 3. Mark McMillin, Public Health Data Science MPH | 2025 - Present |
| 4. Zhiyu Kang, Biostatistics PhD | 2024 - 2026 |
| 5. Rui Cao, Biostatistics PhD | 2024 - 2025 |
| 6. Zheqi Yin, Biostatistics MS | 2024 - 2025 |
| 7. Yuzi Mi, Biostatistics PhD | 2023 - 2025 |
| 8. Tyler Jubenville, Biostatistics MS | 2021 - 2025 |
| 9. Boya Shi, Biostatistics MS | 2022 - 2024 |
| 10. Tianjie Zhang, Biostatistics MS | 2024 |
| 11. Lujun Zhang, Biostatistics PhD | 2022 - 2024 |

Doctoral Dissertation Directed

- | | |
|---|-----------------|
| 1. Josey Sorenson, Biostatistics PhD
Thesis: <i>High-dimensional mediation analysis with applications to multi-omics data</i> | 2026 (expected) |
| 2. Zhiyu Kang, Biostatistics PhD
Co-advised with Dr. Erick Lock
Thesis: <i>High-dimensional Tensor Decomposition and Mediation Analysis</i> | 2026 |
| 3. Rui Cao, Biostatistics PhD
Co-advised with Dr. Weihua Guan
<i>Doctoral Dissertation Fellowship</i>
Thesis: <i>Statistical Models for High-Dimensional Genetic and Genomic Data Analysis</i> | 2025 |

Master's Thesis Directed

- | | |
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| 1. Joshua Hebeisen, Biostatistics MS
Thesis: <i>Building Benchmarking Pipelines for Single-Cell Batch Correction Methods</i> | 2026 |
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Doctoral Final Committee: Committee Chair

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| 1. Michael Anderson, Biostatistics PhD
Thesis: <i>Methods for Supervised Machine Learning and Polygenic Risk Scores</i> | 2024 |
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Doctoral Final Committee: Committee Member

1. Yuting Shan, Exp & Clin Pharmacology PhD 2026
Thesis: *Understanding and Predicting Heterogeneity in Cancer Treatment Response and Adverse Drug Reactions through Multi-omics Approach*
2. Allison Domingues, Epidemiology PhD 2023
Thesis: *Records-based Childhood Cancer Research Applications and Methods*

Doctoral Preliminary Committee: Committee Member

1. Zexi Rao, Biostatistics PhD 2026 - Present
Thesis: *Machine learning for patient-level risk prediction from longitudinal, single-cell, and multimodal omics data*
2. Lilin Wang, Exp & Clin Pharmacology PhD 2026 - Present
Thesis: *Optimizing Cancer Therapy Efficacy and Safety at Single-Cell Resolution*
3. Jialing Liu, Biostatistics PhD 2025 - Present
Thesis: *Bayesian Transfer Learning for High-Dimensional and Longitudinal Biomedical Studies*
Doctoral Dissertation Fellowship
4. Rachel Moss, Bioinform & Comp Biol PhD 2021
Thesis: *Etiology of pediatric solid tumors using multi-omic approaches*

Master's Thesis/Research Committee: Committee Member

1. Boya Shi, Biostatistics MS 2024
Thesis: *Advancing the Analysis of Neurocognitive Functions in Cryptococcal Meningitis Survivors*
2. Seth Bergstedt, Biostatistics MS 2023
Thesis: *Modeling Postoperative Root Canal Therapy Pain using the PREDICT Study: The Predictive Use of Group LASSO Penalized Regression in a Clinical Setting*

Mentoring/Career Advising

1. Jeremy Allred, University of Minnesota Medical School Department of Medicine, Division of Hematology, Oncology and Transplantation K08 Award 2025
2. Xiong Serena, University of Minnesota School of Medicine Department of Family Medicine and Community Health PreK Award at CTSI 2024

Research and Teaching Trainees

1. Josey Sorenson 2025 - 2026
Graduate Research Assistant on Divisional Startup Fund to develop

- new statistical methods related to mediation analysis (PI: Tianzhong Yang)
2. Aparna Srinivasan 2024 - 2025
Graduate Research Assistant evaluating polygenic risk score for childhood cancer survivors (MPI: Cindy Im, Tianzhong Yang)
 3. Josey Sorenson 2024 - 2025
Graduate Research Assistant on genetic discovery related to childhood cancer (PI: Logan Spector)
 4. Liu Yang 2023 - 2025
Graduate Research Assistant on develop new prediction models for childhood cancer survivors (MPI: Cindy Im, Tianzhong Yang)
 5. Josey Sorenson 2021 - 2024
Trainee for T32 Statistical Genetics and Genomics Training program (PI: Saonli Basu)
 6. Rui Cao 2021 - 2024
Graduate Research Assistant on methods development in statistical genetics and genomics (PI: Tianzhong Yang, Weihua Guan)
 7. Michael Anderson 2023
Graduate Research Assistant on developing new statistical methods regarding to interpretable predictive model (PI: Weihua Guan, Tianzhong Yang)
 8. Bhargob Kakoty 2023
Teaching Assistant for PUBH 6450
 9. Emma Billmyer 2023
Teaching Assistant for PUBH 6450
 10. Estaban Lemus Wirtz 2023
Teaching Assistant for PUBH 6450
 11. Eunsan Kim 2023
Teaching Assistant for PUBH 6450
 12. Lujun Zhang 2022 - 2023
Graduate Research Assistant on developing new multi-omics predictive model (PI: Weihua Guan, Tianzhong Yang)
 13. Tiankai Xie 2022 - 2023
Graduate Research Assistant on genetic discovery related to childhood cancer (PI: Logan Spector)
 14. Zhiyu Kang 2022
Graduate Research Assistant on developing new statistical methods related to mediation analysis (PI: Tianzhong Yang)
 15. Lindsay Turner 2021 - 2022
Graduate Research Assistant on CCRF to set up statistical pipelines with application to childhood cancer (PI: Tianzhong Yang)
 16. Haozuo Zhao 2021

Teaching Assistant for PUBH 6450	
17. Linnea York	2021
Teaching Assistant for PUBH 6450	
18. Neelanzana Paddle	2021
Teaching Assistant for PUBH 6450	
19. Zhiyu Kang	2021
Teaching Assistant for PUBH 6450	

SERVICE

Service to the Discipline/Profession/Interdisciplinary Area(s)

Ad Hoc Reviewer

ASA SSGG Student Paper Award Committee	2024 - 2025
Student Paper Award Reviewer for Conference on Statistics in Genomics and Genetics	2025

Chair

Poster Session Chair, Conference on Statistics in Genomics and Genetics	2025
Chair for Challenges and recent advances in high-dimensional mediation analysis, Joint Statistical Meeting	2020
Chair for the New Methods in Statistical Genomics and Genetics Section, Joint Statistical Meeting	2019

Grant Reviewer

National Institute of Health, Cancer Special Emphasis Panel	2025
St. Baldrick Foundation Research Grants	2025
National Institute of Health, Biodata Management and Analysis Study Section	2024

Local Organizing Committee member

STATGEN 2025: Conference on Statistics in Genomics and Genetics	2025
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Member

AACR Award for Outstanding Achievement in Pediatric Cancer Research Committee	2026
AACR Biostatistics in Cancer Research Task Force	2026

Ad Hoc Reviewer

Bioinformatics
Biostatistics
BMC Bioinformatics
BMC Medical Genetics
Cancer Epidemiology

Cancer Research Communications
Computational Statistics and Data Analysis
Frontiers in Genetics
Genetic Epidemiology
Genetics
HGG Advances
Human Genetics
Journal of Biopharmaceutical Statistics
Nature Communications
Nature Genetics
PLOS Computational Biology
PLOS Genetics
PLOS One
RMD Open
Statistical Analysis and data mining
Statistics in Biosciences
Statistics in Medicine

Service to the University/College/Department

University of Minnesota

Member, Biostatistics and Health Data Science Seminar Committee	2025 - Present
Judge, SPH Research Day Poster	2025
Member, Biostatistics and Health Data Science Recruiting Committee	2021 - 2025
Reviewer, Doctoral Dissertation Fellowship Review Committee	2023
Member, Biostatistics and Health Data Science Exam Committee	2020- Present
Member, Biostatistics and Health Data Science Student Awards Committee	2020

Masonic Cancer Center

Member, Data Safety and Monitoring Committee	2025 - Present
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